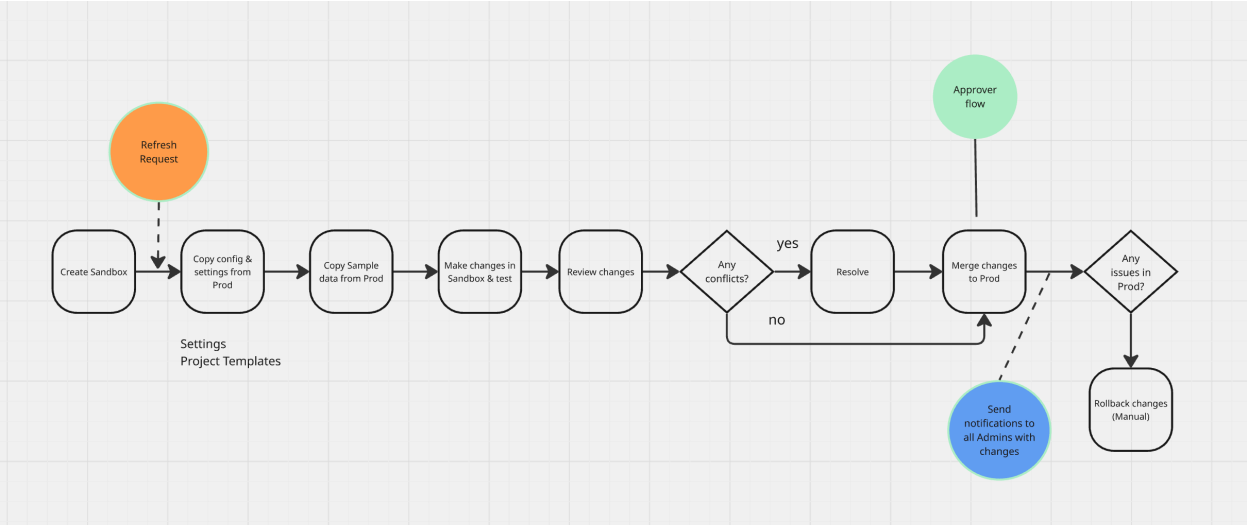
Rocketlane Sandbox



1. What & Why

Sandbox: Isolated testing environment for safe configuration testing before production deployment.

Core Capabilities:

- Create sandbox (1-click copy: all configs + '30' days data)
- Test changes in isolation
- Review changes (**New/Modified/Deleted**)
- Resolve conflicts (manual 3-way merge)
- Merge to production (selective, transaction-safe)
- Audit trail (90-day change log + forever activity log)

URL: {company}-{sandbox-slug}.rocketlane.com

Current Pain:

- Manual workaround exists: External partners copy prod → trial → test → manual merge (2-3 days, error prone)
- CSMs sometimes assist with this manual process
- No conflict detection when prod changes during testing
- No audit trail of what changed
- Production incidents from untested config changes

Business Value:

- Validated demand: Partners/CSMs already doing this manually
- ROI: Save 2-3 days per implementation, remove manual errors.
- Competitive parity: All enterprise platforms have sandbox (Salesforce, NetSuite, ServiceNow, HubSpot, Workday etc)

2. Market Research

Industry Standard: Manual Conflict Resolution

Platform	Conflict Resolution	Auto-Merge	Native Rollback	
Salesforce	3-way merge (field-level)	Non-overlapping only	NO	
NetSuite	Accept or Skip	None	NO	
ServiceNow	Per-collision (Accept or Skip)	None	NO	

Key Findings:

1. Sandbox is table stakes - all enterprise platforms offer it

2. Manual conflict resolution is standard - No competitive disadvantage
3. No automatic rollback - Industry accepts detailed audit logs for manual rollback
4. Third-party opportunity - Rollback gap created profitable market (Phase 2 potential)

3. Functional Requirements

Step 1 - Sandbox Creation

Trigger: Production > Settings > Sandboxes > “Create Sandbox”

> 💡 **Creation Failure Handling:** Sandbox is not accessible until creation is 100% complete. On failure, the system auto-purges the partial instance, auto-retries up to ‘x’ times (need to finalise the no of time after discussing with tech team). If all retries fail, admins are notified via email and and the "Create Sandbox" button in Settings > Sandboxes becomes available again for manual retry.

Requirements:

- Permission: "Manage Sandboxes" (includes external partners with delegated customer access)
- Tier limits:
  - Team: Not available
  - Professional: 0 included (must purchase if needed)
  - Enterprise:
    - 1 sandbox included FREE (without data)
    - For Sample data - we can charge 5% of ACV
    - For additional sandboxes - 10% of ACV?
- Async process

- Communicate approx. time line → ex - 2 hours for creation (will be dynamic and will change based on volume of data to be copied)
- Email admins when sandbox is ready
- URL: {company}-{sandbox-slug}.rocketlane.com

Step 2 - Data Copy

What gets copied completely:

1. Templates
2. Automations
3. Dashboard/Report Definitions - configs copied fully, data displayed is based on 30-day window
4. Custom Reports - configs copied fully, data displayed is based on 30-day window
5. Custom Apps
6. Settings
    a. Personal
    b. Customer Portal
    c. Account (except Partners, Secret Store)
    d. AI capabilities
    e. Operations & Financials
    f. Sprints
    g. Fields
    h. Workflow Management

What gets copied partially

1. Accounts
2. Projects → All linked data like Phases, allocations, time entries

3. Tasks
4. Expenses
5. Invoices
6. Revenue Rec ledger
7. Sprint planning

What NOT to copy

1. Archived Projects
2. API keys
3. Webhooks
4. Secret Store
5. SCIM
6. Integrations

Data (30 Days Only):

- Projects (created_at or updated_at >= TODAY - 30 days) + linked data (phases, tasks, allocations, time entries). All projects which are created/updated in the last 30 days.
- Accounts
- Time entries, Expenses, Invoices (30 days, linked to projects)
- Documents (max 100 per project)
- Team Members →
  - All admins,
  - team members linked to Automations
  - team members linked to the projects - covers task assignees, phase owners, allocation resources within the 30-day window

Sandbox Isolation

Requirements:

- Separate database instances
- Changes don’t affect production
- Integrations: “Not Connected” status (must reconnect with TEST credentials)
- Emails: Logged but NOT sent (prevent customer emails)
- Visual: Clear “SANDBOX” banner on every page

Step 3 - Review Changes (Compare Sandbox vs Production)

Trigger: Sandbox > “Sync to Production” > Review Changes

Detection:

- Compare Day 1 snapshot Vs Current Production Vs Current Sandbox. Day 1 snapshot will live as long as the sandbox. Snapshot expires only when the sandbox is deleted, refreshed, or merged.
- **Cross-sandbox invalidation:** If production changes while this sandbox's Review Changes screen is open, either from a direct prod edit or from another sandbox merging to prod, then show: "Production has changed since you opened this review. Click 'Refresh' to see latest diff." (This applicable ONLY when a customer uses Multiple Sandboxes)
- Categorise: NEW, MODIFIED, DELETED

NEW objects:

- User just built this in sandbox
- User knows what it is (they created it)
- No comparison needed (nothing in production to compare)
- Just preview: "Here's what you're adding"

How to view changes

- Opens the object in a new tab

- URL: `https://acme-sandbox.rocketlane.com/workflows/client-onboarding-v2`
- User sees the full configured object
- **Important:** Opens in SANDBOX (where it exists), not production

MODIFIED objects:

- User changed something
- Production has old version
- **Critical to show:** What changed?
- This is where diff/comparison matters

How to View changes

- For simple changes: Expands inline diff
- For medium changes: Opens modal comparison
- For complex changes: Downloads comparison file

DELETED objects:

- Production has it
- Sandbox doesn't
- **Critical to show:** What's being removed + impact

How to View changes

- Shows what's being removed
- Shows where it's currently used
- Warns about impact

> 💡 If a user modifies Production during "Review Changes". Add - "Production has changed since you opened this review. Click 'Refresh' to see latest diff."

Display:

Change Type	Display	Default	Action
NEW	Object type + name + stats	Checked	User can uncheck
MODIFIED	Side-by-side diff (inline/modal/Excel based on complexity)	Checked	User can uncheck
DELETED	Object name + impact warning	Unchecked	User must explicitly check

Tiered Viewing:

- Simple (<10 changes): Inline expansion
- Medium (10-50): Modal comparison
- Complex (50+): Excel export (4 sheets: Summary, Added, Modified, Deleted)

Validation:

- Breaking changes: Detect field type changes, required field additions → Show impact
- Dependencies: Block merge if missing dependencies → Clear error message

Selective Merge: Checkboxes allow deploying subset of changes

Market Comparison:

Platform	Approach	Granularity
Salesforce	Field-level comparison, colour highlighting	Field-level
NetSuite	Object-level preview, red highlighting	Object-level

Step 4 - Conflict Resolution (3-Way Merge)

Trigger: Review Changes detects conflict

Detection Logic:

- Compare: Day 1 snapshot + Current Production + Current Sandbox
- Conflict = Same object modified in BOTH production AND sandbox since Day 1

UI:

- [ ] Resolve Conflicts screen
- [ ] Side-by-side comparison view
- [ ] Radio buttons for choice
- [ ] Acknowledgment checkbox when overwriting
- [ ] Progress indicator
- [ ] Block "Continue" until all resolved

Resolution flow:

Conflict 1 of 3: Automation “Project Kickoff” - Step 3

Production (Modified by Admin A, 5 days ago):
Step 3: Send email to client

Sandbox (Modified by Admin B, today):
Step 3: Send Slack to PM

- Keep Production (discard sandbox changes)
- Keep Sandbox (overwrite Admin A’s changes)
  [If Keep Sandbox selected:]
  ✓ I acknowledge this will overwrite Admin A’s changes
  ✓ Notify Admin A

Session Persistence: Conflict resolution choices are saved server-side automatically as the user progresses. On returning to the Resolve Conflicts screen, previously resolved conflicts show their saved choice and are collapsed

by default. User can continue from where they left off or revisit any resolved conflict to change their choice before hitting Merge.

Step 5 - Merge to Production

Trigger: Review Changes > Resolve Conflicts > “Merge”

Process:

1. Summary: Count of changes by type
2. Transaction-safe execution

Merge Transaction flow:

1. Begin Merge
2. For each change:

   Write Attempt Log → [outside transaction]
   Begin Transaction
   → Apply change to production
   → Log change detail to audit table
   → If ANY fails: ROLLBACK
   Commit
   Write Outcome Log → [outside transaction, success or failure]
3. If ANY fails:
   - ROLLBACK entire transaction
   - Send alert to admins
   - Preserve sandbox state
4. COMMIT only if all succeed
5. Show all the progress visually to the users.
6. Sandbox can not be edited till the merge is completed.
7. Email notification

After a successful merge, Sandbox persists as-is (now in sync with prod, but snapshot is gone)

Audit Log

- Merge change log: 90-day retention (before/after values, field diffs, rollback instructions)
- Activity audit log: Forever retention (all sandbox activities for compliance)

Rollback Support (Manual):

- For each change, provide instruction:
- Added: “Delete [Object] from Production”
- Modified: “Change [field] from [new] back to [old]”
- Deleted: “Recreate [Object] with definition: [JSON]”
- Excel export with rollback instructions
- “Copy Before Value” buttons

Market Reality: 0% of platforms have automatic rollback → Manual with audit logs is standard

Step 6 - Sandbox Refresh

Trigger: Production > Settings > Sandboxes > “Refresh”
Warning: “This will PERMANENTLY DELETE all sandbox work that hasn’t been merged”

Manual Refresh Flow

- [ ] "Refresh Sandbox" button in settings
- [ ] Confirmation dialog
- [ ] Email notification (sent to all admins that a refresh has started)
- [ ] Show progress
- [ ] Sandbox updated with prod data

- [ ] Email notification when complete (sent to all admins)
- [ ] Error handling & retry logic
- [ ] Audit logging

Step 7 - Activity Audit Log

Forever retention for compliance:

Use Cases:

- Compliance reporting (SOC2, GDPR)
- Usage analytics (adoption, merge frequency, conflict patterns)
- Attribution (who changed what in production, trace to sandbox)

Step 8 - High-Level Architecture

Components:

1. Sandbox Manager: Lifecycle (create, refresh, delete), metadata
2. Data Copy Service: Batch operations, parallel processing, progress tracking
3. Change Detection: Compare sandbox vs production, detect conflicts (3-way)
4. Merge Service: Transaction management, atomic operations, auto rollback on failure
5. Audit Service: Change logs (90 days), activity logs (forever), query API

RBAC

Two new sandbox-specific permissions:

1. Manage Sandboxes

- Create sandbox
- Refresh sandbox

- Delete sandbox
- View all sandboxes in the org

Only admins + external partners with delegated access should have this.

2. Merge to Production

- Initiate sync to production
- Resolve conflicts
- Review and select changes

Inside the Sandbox

Once someone has access to a sandbox, what can they do within it?

Mirror production roles
Whatever role a user has in production, they get the same in sandbox. Simple, consistent, no extra management overhead.

External Partners Specifically

- Can be granted Manage Sandboxes explicitly by a customer admin
- Should NOT have Merge to Production by default, this should be a conscious grant
- Losing production access = losing sandbox access immediately (as already resolved)

What Production Admins Always Retain

Regardless of who created the sandbox or who's been working in it:

- Full visibility into all sandboxes in the org
- Ability to delete any sandbox
- Ability to merge any sandbox to production
- Access to full audit log

This ensures no sandbox ever becomes a blind spot for the customer's own team.

Summary Table

Action	Internal Admin	Internal Member	External Partner
Create Sandbox	✅	❌	✅ (if granted)
View Sandbox	✅	✅ (if granted)	✅ (if granted)
Work in Sandbox	✅	✅ (if granted)	✅ (if granted)
Refresh Sandbox	✅	❌	❌
Merge to Production	✅	❌	✅ (if explicitly granted)
Delete Sandbox	✅	❌	❌
View Audit Log	✅	❌	❌

Edge Cases

1. **External Partner Access After Revocation**

Partner access to sandbox is derived from production access. Deactivating a partner immediately revokes their sandbox access. The sandbox itself persists untouched, it was always owned by the customer org. All admins are notified: *"Partner X has been deactivated and no longer has sandbox access."* Any admin with "Manage Sandboxes" permission can pick up and complete any pending work.

2. **Selective merge + dependencies could deadlock**

   Use case -

   - User creates a new **Automation** in sandbox that references a new **Custom Field** they also created
   - They select only the Automation to merge, not the Custom Field
   - Automation lands in production referencing a field that doesn't exist there
   - System blocks it but now the user is stuck with no clear path forward

   Solution -

   Before the user even selects what to merge, the system should visually indicate coupled changes. Something like:

   - A small link icon between dependent objects

- Hovering shows: *"This automation requires Custom Field X to be merged together"*
- If user unchecks one half of a dependency pair, the other gets a warning indicator

This is the most user-friendly approach, we are preventing the deadlock rather than reacting to it.

Fallback: If they still try to merge with broken dependencies

"2 of your selected changes have missing dependencies. Here's what's needed:"

- Custom Field "Client Tier" - required by Automation "Project Kickoff" → **[Add to merge]**
- Workflow Stage "Review" - required by Template "Onboarding v2" → **[Add to merge]**

One-click **Add to merge** per dependency. Plus a **"Add all missing dependencies"** bulk action.

3. **Circular Dependencies - TBD**

Deferred (Will pick up based on customer request)

Configurable Data Copy Range

- Current: Fixed 30 days
- Phase 2: 90 days, 180 days, full data options. Option to copy specific projects etc.

Approval Workflow

- Current: Direct merge
- Phase 2: Requester → Approver → Merge
- Rationale: Multiple checkpoints exist, not critical

Others

- Multiple sandbox types (Developer, Developer Pro, Partial, Full) - similar to Salesforce
- Cross-sandbox merging (Sandbox A → Sandbox B)
- Version control integration (Git)

Risks & Mitigation

- Sandbox creation takes too much time
- Merge failures corrupt production (High)
- Conflict detection misses Edge Cases (Medium)

Appendix

Summary Pricing Table with Sources

Platform	Free Sandbox with Enterprise?	Pricing Details	Primary Source
HubSpot	**YES - 1 FREE** with ALL Enterprise Hubs	**FREE:** 1 standard sandbox **PAID:** $750/month per additional sandbox	HubSpot Official
Salesforce	**Developer Only** (metadata, no data)	**FREE:** Developer Sandbox (metadata only) **PAID:** 1. Developer Pro: 5% of net spend (~$25-300/mo) 2. Partial Copy: 20% of net spend (~$300-3K/mo) 3. Full Copy: 30% of net spend (~$1.5K-15K+/mo)	Arrify GuideVendr Community
NetSuite	**Some editions include**	**VARIES BY EDITION:**• Some packages bundle sandbox **IF NOT INCLUDED:**• Standard: 10% of license (~$1K-10K/year)• Premium: 20% of license (~$2K-20K/year)	NoBlue2 GuideHouseblend Analysis

Platform	Free Sandbox with Enterprise?	Pricing Details	Primary Source
ServiceNow	**Likely included** in Enterprise packages	**NOT PUBLICLY DISCLOSED**• All pricing is custom quote-based• Test environments likely included in higher tiers• Additional environments = custom pricing	Hiverhq AnalysisEesel AI Guide
Workday	**YES - 2 FREE** with ALL plans	**FREE (INCLUDED):**• Production tenant• Sandbox (weekly refresh)• Sandbox Preview (bi-annual refresh)**PAID:**• Implementation tenants (on-demand refresh, pricing not disclosed)	Workday DatasheetCustomer Blog

1. HubSpot

Pricing: 1 FREE + $750/month for additional

Primary Source:

- **HubSpot Official Product Page**
  - URL: https://www.hubspot.com/products/sandbox
  - Quote: *"Each account includes one sandbox, with more available for purchase for $750/sandbox/month."*

2. Salesforce

Pricing: % of Net Spend (5%, 20%, 30%)

Primary Sources:

- **Arrify Salesforce Sandbox Guide** (October 2025)
  - URL: https://arrify.com/salesforce-sandbox

- Quote: *"Developer Sandbox: It's free...Developer Pro Sandbox: It costs 5% of net spend. Partial Copy Sandbox: It costs 20% of net spend. Full Copy Sandbox: It costs 30% of net spend."*

- **Vendr Salesforce Pricing Community**
  - URL: https://www.vendr.com/marketplace/salesforce
  - Quote: *"The Sandbox pricing is tiered as a percentage of your net software spend. For Developer Sandbox users, it's complimentary; for Developer Pro, expect to add 5%; and for more feature-rich environments like Partial Copy and Full Copy, the costs are 20% and 30% respectively."*
- **Prodly Sandbox Cost Savings Blog**
  - URL: https://www.prodly.co/blog/5-tips-to-save-on-sandbox-costs
  - Quote: *"A Full Copy sandbox costs 30% of your net spend on your Salesforce license. For this money, you get a complete replica of your production org with all the customer data and metadata."*

Example Calculations:

- **"Things You Don't Need: A Full Sandbox"** (January 2024)
  - URL: https://www.freelikeapuppy.tech/post/things-you-usually-don-t-need-a-full-sandbox
  - Quote: *"a full copy sandbox costs 30% of the net licensing cost of your org...the minimum cost of a full copy sandbox is $1,485."*

Official Salesforce:

- **Platform Sandboxes Pricing Page**
  - URL: https://www.salesforce.com/platform/sandboxes-environments/pricing/
  - Note: Page provides minimal detail, states pricing is based on percentage of applicable products

3. NetSuite

Pricing: 10% (Standard) / 20% (Premium) of Annual License

Primary Sources:

- **NoBlue2 NetSuite Sandbox Guide** (September 2024)
  - URL: https://noblue2.com/blog/resource-netsuite-sandbox/
  - Quote: *"The price of the NetSuite Sandbox is calculated at 10% of the sum total of the NetSuite modules and users purchased. The NetSuite premium sandbox is calculated at 20% of the sum total of the NetSuite modules and users purchased."*
- **Houseblend NetSuite Licensing Guide**
  - URL: https://www.houseblend.io/articles/netsuite-licensing-pricing-structure-erp
  - Quote: *"Additional Environments (Sandbox): ...your main production subscription cost). For example, a sandbox might cost roughly 10% of your base fee....Include that cost if relevant (e.g. if your annual NetSuite is $100k, a sandbox might be ~$10k/year)."*
- **Closeloop NetSuite Pricing Guide** (June 2025)
  - URL: https://closeloop.com/blog/netsuite-pricing-and-implementation-costs/
  - Quote: *"For companies that need a safe space to test configurations, customizations, or integrations without impacting live data, NetSuite offers Sandbox accounts...Sandbox environments are not included by default and are priced as add-ons...Most mid-sized companies should expect to pay several thousand dollars annually per sandbox."*

Community Insights:

- **NetSuite Professionals Community** (November 2025)
  - URL: https://archive.netsuiteprofessionals.com/t/32723095/how-much-does-a-sandbox-typically-cost
  - Quote: *"The cost of a sandbox in NetSuite typically varies as it is usually a percentage of your license cost. From the conversations, it appears that the cost can range from 10% to 20% of the license cost."*

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4. ServiceNow

Pricing: Not Publicly Disclosed

Primary Sources:

- **Hiverhq ServiceNow Pricing Guide** (November 2025)
  - URL: https://hiverhq.com/blog/servicenow-pricing
  - Quote: *"ServiceNow does not publish standard pricing, uses a mix of license types, bundles modules separately, and treats AI usage, implementation, and renewals as additional cost layers."*
- **Eesel AI ServiceNow Pricing Analysis**
  - URL: https://www.eesel.ai/blog/servicenow-pricing
  - Quote: *"ServiceNow does not just list their prices online for everyone to see. Instead, they build a custom quote for each customer based on a few specific factors."*
- **SmartSuite ServiceNow Pricing** (December 2025)
  - URL: https://www.smartsuite.com/blog/servicenow-pricing
  - Quote: *"ServiceNow's pricing plans are not publicly listed, as they offer consultation-based pricing through a 1:1 sales call...Based on ServiceNow customer and public reviews, the average cost of ServiceNow contracts is $130,080 annually."*

Note: No specific sandbox pricing found because ServiceNow doesn't publish any pricing publicly. Everything is custom-quoted.

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5. Workday

Pricing: FREE (Included) + Paid Implementation Tenants

Primary Sources:

- **Workday Official Datasheet** (2025)
  - URL: https://www.workday.com/content/dam/web/en-us/documents/datasheets/workday-tenants-datasheet.pdf

- Quote: *"Production tenant. The Sandbox...60 days after deployment has occurred, customers commonly purchase multiple implementation tenants. The implementation tenant is a direct replica of the production environment. It is refreshed in an on-demand fashion."*

- **PeerSpot Community Review**
  - URL: https://www.peerspot.com/questions/what-is-your-experience-regarding-pricing-and-costs-for-workday
  - Quote: *"Workday, by default, is going to give you production and a sandbox environment with your license. They're also going to give you a preview environment of what's being released, but if you want additional test environments, there is a license cost."*
- **The Wonders of Workday Blog** (August 2024)
  - URL: https://www.customersharingmovement.com/post/the-wonders-of-workday
  - Quote: *"Workday's standard setup comes with both a Sandbox and a Sandbox Preview tenant. Sandbox refreshes every Friday night, and Sandbox Preview refreshes twice a year before each major release. This is incredibly valuable, especially since it's included at no extra cost. Other vendors charge a hefty fee for a test environment and limit the number of refreshes."*
- **Surety Systems Workday Guide** (June 2025)
  - URL: https://www.suretysystems.com/insights/your-guide-to-navigating-the-workday-sandbox/
  - Quote: *"Sandbox tenants are one of three types of production tenants. The main goal is to test and prep modifications before actually implementing them in production tenants."*

Third-Party Rollback Market

Tool	Platform	Price	Features
Gearset	Salesforce	$100-$300/user/mo	Auto snapshots, one-click rollback, partial rollback

Tool	Platform	Price	Features
Rewind	HubSpot, Shopify	$49-$99/mo	Backup & restore
Salto	NetSuite, Salesforce	Custom pricing	Version control, rollback

Implication: Customers pay $\$50 - \$300/$ month for rollback → Phase 2 opportunity

Open Questions

1. Exact pricing for additional sandboxes?
2. RBAC
3. While copying sample data → will there by any PII data issue → Need broader discussion
4. What happens to sandboxes that are:
   - Created 6 months ago
   - Never merged
   - Never accessed
   - Taking up storage

   Can we do Auto-Cleanup:

   - Warn admins (All admins with Manage Sandboxes permission) at 60 days of inactivity (no users logged in to Sandbox)
   - Warn again at 80 days
   - Auto-delete at 90 days (with final warning email)
   - User can mark as "Keep Forever" to prevent deletion. Only admins with “Manage Sandbox permission” should be able to do it.
5. Production Deleted While Sandbox Exists - When production is deleted, sandbox also will be deleted (will happen only when a customer churns)
6. User Modifies Production During "Review Changes". Add -

"Production has changed since you opened this review.
Click 'Refresh' to see latest diff."

Resources

- Figma Designs -